

DEVELOPING PCBs USING ExpressPCB

Here's a simple way to make your own PCBs. All the required components are readily available at electronics hardware shops

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As an electronics engineering student myself, I know how difficult it is to assemble electronic circuits on Vero boards. You have to connect each and every component using wires, which most of the times break apart or cut loose. Wiring is also very cumbersome.

Printed circuit boards (PCBs), which are specifically made for each circuit, come in handy. These are very compact and wire-free. But for a normal hobbyist or student, purchasing a PCB from the shop for a specific circuit can be very expensive. So I searched for a way of making my own PCB in a simple way. Finally, I succeeded and have been making my own PCBs for two years now. I hope you will find this method interesting as it gives you the freedom to make

your own customised PCBs. All the required components are readily available at electronics hardware stores. So why wait; get started now.

Five steps to PCB making

1. Draw the schematic of the circuit on a computer using the required software
2. Design the PCB on the computer using the required software
3. Print the PCB design through a laser printer
4. Take the impression of the circuit on a copper-clad board
5. Remove the excess copper by etching

Drawing the schematic

There are many software available on the Internet to help you draw the schematic of the circuit. Eagle software is a very good option but it is not a free-

ware. So here we will use ExpressSCH for drawing the schematic. ExpressPCB (which includes ExpressSCH) is available on the Internet for free.

Circuit description. Let us assemble a simple buffer circuit using the 74HC244 buffer IC. We will put a tap switch at one input and see that we get the same output as the input by putting an LED light at the output pin. So when we push the switch to 'on' position the LED will glow, and when we push it to 'off' position the LED will turn off.

Fig. 1 shows the screenshot of the buffer circuit drawn in ExpressSCH software.

Screenshot description. Icons marked 1 through 24 allow you to perform various tasks as follows:

1. Rotate the component as well as the text by 90°. It can also be used to rotate only the text

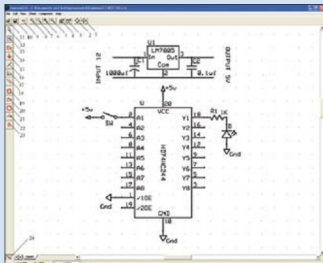


Fig. 1: Screenshot of the buffer circuit drawn in ExpressSCH software

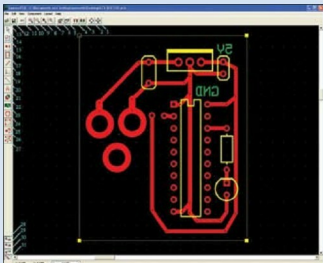


Fig. 2: PCB of the schematic (shown in Fig. 1) designed in ExpressPCB